

Dynamic Analysis of a Quarter-Car Suspension System Using MATLAB/Simulink Under Sinusoidal Road Excitation

A Simulation-Based Engineering Study

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Abstract

Vehicle suspension systems play an important role in maintaining ride comfort, controlling wheel motion, and ensuring effective interaction between the tyre and road surface. The dynamic behaviour of a suspension system depends on parameters such as sprung mass, unsprung mass, suspension stiffness, damping coefficient, tyre stiffness, and road excitation frequency.

In this work, a linear two-degree-of-freedom quarter-car suspension model is developed to investigate the vertical dynamic response of a vehicle subjected to a sinusoidal road profile. The model consists of a sprung mass representing one quarter of the vehicle body and an unsprung mass representing the wheel and associated components. The suspension spring and viscous damper are placed between the sprung and unsprung masses, while the tyre is represented by a linear stiffness element between the wheel and road.

The governing equations of motion are derived using Newton's second law and implemented in both MATLAB and Simulink. The baseline model uses a sprung mass of 300 kg, unsprung mass of 40 kg, suspension stiffness of 15 000 N/m, suspension damping coefficient of 1500 N s/m, and tyre stiffness of 150 000 N/m. A sinusoidal road input with an amplitude of 0.05 m and wavelength of 5 m is considered.

Parametric studies are performed by varying the suspension damping coefficient and vehicle velocity. The damping study is conducted at a vehicle velocity of 5 m/s, while the velocity study considers velocities of 5 m/s, 10 m/s, and 15 m/s. Body displacement, suspension deflection, and wheel displacement are used as primary response measures. The results show that increasing the damping coefficient from 500 N s/m to 1500 N s/m reduces the maximum body displacement from 0.112 30 m to 0.064 63 m. For the velocity study, the maximum body displacement decreases from 0.064 63 m at 5 m/s to 0.013 32 m at 15 m/s, while the maximum wheel displacement increases from 0.005 608 m to 0.011 93 m.

The study demonstrates how a relatively simple quarter-car model can be used to investigate the influence of suspension and operating parameters on vehicle vertical dynamics.

Keywords: Quarter-car model, vehicle suspension, sprung mass, unsprung mass, suspension damping, tyre stiffness, road excitation, MATLAB, Simulink, vehicle dynamics.

1 Introduction

A vehicle suspension system connects the vehicle body to the wheels and plays a major role in controlling the vertical motion of the vehicle. When a vehicle travels over an uneven road, the road profile produces vertical excitation at the tyre. This excitation is transmitted through the wheel and suspension system to the vehicle body. The resulting body motion directly influences ride comfort, while the wheel motion affects tyre–road contact and vehicle handling.

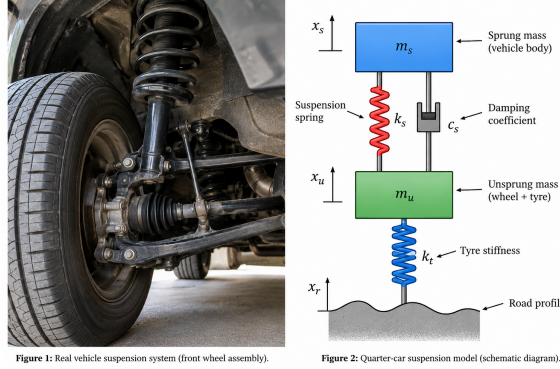


Figure 1: Real automotive suspension assembly and its simplified quarter-car representation.

A suspension system generally consists of springs, dampers, tyres, and associated mechanical components. The spring provides elastic support and stores mechanical energy, whereas the damper dissipates energy and reduces oscillatory motion. The tyre also contributes significant vertical stiffness and therefore affects the dynamic response of the wheel assembly.

The complete vehicle suspension system is a complex multi-body system involving vertical motion, pitch, roll, suspension geometry, tyre dynamics, and interactions between different wheels. For preliminary analysis and parameter studies, however, the system can be simplified using a quarter-car model.

The quarter-car model represents one quarter of the vehicle and considers the vertical motion of the vehicle body and one wheel assembly. In its commonly used two-degree-of-freedom form, the model contains a sprung mass and an unsprung mass. The sprung mass represents the corresponding portion of the vehicle body, while the unsprung mass represents the wheel, tyre, and associated components.

The present study develops a mathematical model of a two-degree-of-freedom quarter-car suspension system and investigates its response to a sinusoidal road input. The model is implemented using MATLAB and Simulink so that both numerical simulation and block-diagram-based modelling can be used to study the system.

Particular attention is given to the influence of suspension damping and vehicle velocity. The damping coefficient is varied to investigate its influence on body motion,

suspension deflection, and wheel displacement. Vehicle velocity is varied while maintaining the road wavelength constant, thereby changing the frequency of road excitation experienced by the vehicle.

The main objectives of the study are:

1. To develop a mathematical model of a two-degree-of-freedom quarter-car suspension system.
2. To derive the governing equations of motion for the sprung and unsprung masses.
3. To model a sinusoidal road excitation as a function of vehicle velocity.
4. To implement the mathematical model in MATLAB using numerical integration.
5. To develop an equivalent Simulink block-diagram model.
6. To investigate the influence of suspension damping on the dynamic response.
7. To investigate the influence of vehicle velocity on the dynamic response.
8. To compare body displacement, wheel displacement, and suspension deflection for different operating conditions.

2 Quarter-Car Suspension Model

2.1 Physical Representation

The quarter-car model considered in this study is a two-degree-of-freedom vertical suspension model. It represents one wheel assembly and the corresponding quarter of the vehicle body.

The physical arrangement consists of a sprung mass m_s , an unsprung mass m_u , a suspension spring k_s , a suspension damper c_s , and a tyre stiffness k_t .

The sprung mass is connected to the unsprung mass through the suspension spring and damper. The unsprung mass is connected to the road through the tyre stiffness.

The vertical displacements are defined as follows:

- $x_s(t)$: vertical displacement of the sprung mass,
- $x_u(t)$: vertical displacement of the unsprung mass,
- $x_r(t)$: vertical displacement of the road surface.

The corresponding physical chain is

Road $\rightarrow$ Tyre $\rightarrow$ Unsprung mass $\rightarrow$ Suspension $\rightarrow$ Sprung mass.

The suspension spring and damper act in parallel between the sprung and unsprung masses.

2.2 Physical Model Diagram

A schematic representation of the quarter-car system is shown in Fig. 2.

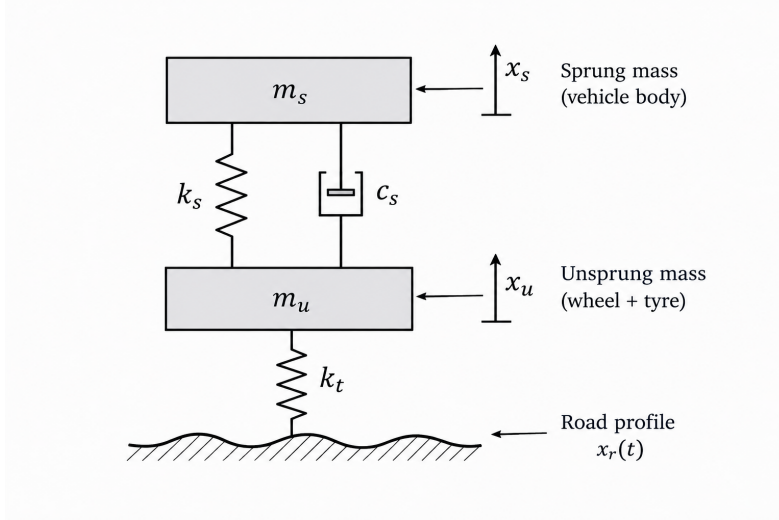


Figure 2: Two-degree-of-freedom quarter-car suspension model.

3 Model Assumptions

The following assumptions are made to obtain a linear and computationally manageable quarter-car model:

1. The complete vehicle is represented by one quarter of the vehicle, consisting of one wheel assembly and the corresponding portion of the vehicle body.
2. The system has two degrees of freedom corresponding to the vertical displacements of the sprung and unsprung masses.
3. Only vertical motion is considered. Longitudinal and lateral vehicle motions are neglected.
4. Vehicle pitch and roll motions are neglected.
5. The vehicle travels at a constant forward velocity during each simulation.

6. The road surface is represented by a prescribed sinusoidal profile.
7. The suspension spring is assumed to be linear with constant stiffness k_s .
8. The suspension damper is assumed to be a linear viscous damper with constant damping coefficient c_s .
9. The tyre is represented by a linear vertical stiffness k_t .
10. Tyre damping is neglected.
11. Nonlinear suspension effects such as bump stops, clearances, progressive springs, and nonlinear damper characteristics are neglected.
12. Friction and other secondary dissipative effects are neglected.
13. The sprung and unsprung components are represented as rigid lumped masses.
14. The masses of the suspension spring and damper are neglected.
15. The model is linearized about the static equilibrium position.
16. The tyre remains in contact with the road throughout the simulation.
17. Gravitational and static equilibrium forces are omitted from the dynamic equations because they cancel when the equations are written about the equilibrium position.
18. Aerodynamic forces are neglected.
19. Detailed suspension geometry and kinematic effects are not represented.
20. Interaction between the four wheels is not explicitly modelled; the quarter-car system is considered independently.

4 Governing Equations of Motion

4.1 Suspension Spring and Damper Forces

The relative displacement across the suspension is

$$x_s - x_u.$$

Therefore, the suspension spring force is

$$F_s = k_s(x_s - x_u). \tag{1}$$

The relative velocity across the damper is

$$\dot{x}_s - \dot{x}_u.$$

Hence, the damping force is

$$F_d = c_s(\dot{x}_s - \dot{x}_u). \quad (2)$$

The total suspension force is therefore

$$F_{\text{susp}} = k_s(x_s - x_u) + c_s(\dot{x}_s - \dot{x}_u). \quad (3)$$

4.2 Tyre Force

The tyre deformation relative to the road is

$$x_u - x_r.$$

The tyre force is therefore

$$F_t = k_t(x_u - x_r). \quad (4)$$

4.3 Equation of Motion of Sprung Mass

Applying Newton's second law to the sprung mass gives

$$m_s \ddot{x}_s = -k_s(x_s - x_u) - c_s(\dot{x}_s - \dot{x}_u).$$

Therefore,

$$m_s \ddot{x}_s + c_s(\dot{x}_s - \dot{x}_u) + k_s(x_s - x_u) = 0. \quad (5)$$

The acceleration of the sprung mass can be written as

$$\ddot{x}_s = \frac{-k_s(x_s - x_u) - c_s(\dot{x}_s - \dot{x}_u)}{m_s}. \quad (6)$$

4.4 Equation of Motion of Unsprung Mass

For the unsprung mass, the suspension force acts in the opposite direction to that on the sprung mass, while the tyre provides the road reaction.

Applying Newton's second law gives

$$m_u \ddot{x}_u = k_s(x_s - x_u) + c_s(\dot{x}_s - \dot{x}_u) - k_t(x_u - x_r).$$

Therefore,

$$m_u \ddot{x}_u = k_s(x_s - x_u) + c_s(\dot{x}_s - \dot{x}_u) - k_t(x_u - x_r). \quad (7)$$

The corresponding acceleration is

$$\ddot{x}_u = \frac{k_s(x_s - x_u) + c_s(\dot{x}_s - \dot{x}_u) - k_t(x_u - x_r)}{m_u}. \quad (8)$$

5 Road Excitation Model

The road surface is represented by a sinusoidal profile. If X represents the longitudinal position of the vehicle, the road displacement is defined as

$$x_r(X) = A \sin\left(\frac{2\pi X}{\lambda}\right), \quad (9)$$

where

- A is the road amplitude,
- λ is the road wavelength.

For a vehicle travelling at constant velocity v ,

$$X = vt. \quad (10)$$

Therefore, the road displacement becomes a function of time:

$$x_r(t) = A \sin\left(\frac{2\pi vt}{\lambda}\right). \quad (11)$$

The corresponding excitation frequency is

$$f = \frac{v}{\lambda}. \quad (12)$$

The angular frequency is

$$\omega = 2\pi f = 2\pi \frac{v}{\lambda}. \quad (13)$$

Thus, the road input can also be written as

$$x_r(t) = A \sin(\omega t). \quad (14)$$

For the baseline condition,

$$A = 0.05 \text{ m}, \quad \lambda = 5 \text{ m}, \quad v = 5 \text{ m/s}.$$

Therefore,

$$f = \frac{5}{5} = 1 \text{ Hz}, \quad (15)$$

and

$$\omega = 2\pi(1) = 2\pi \approx 6.283 \text{ rad/s}. \quad (16)$$

6 State-Space Representation

For numerical simulation, the second-order equations are converted into a set of first-order differential equations.

The state vector is defined as

$$\mathbf{X} = \begin{bmatrix} x_s & \dot{x}_s & x_u & \dot{x}_u \end{bmatrix}^T. \quad (17)$$

The corresponding state equations are

$$\dot{x}_1 = x_2, \quad (18)$$

$$\dot{x}_2 = \frac{-k_s(x_1 - x_3) - c_s(x_2 - x_4)}{m_s}, \quad (19)$$

$$\dot{x}_3 = x_4, \quad (20)$$

and

$$\dot{x}_4 = \frac{k_s(x_1 - x_3) + c_s(x_2 - x_4) - k_t(x_3 - x_r)}{m_u}. \quad (21)$$

Thus,

$$\dot{\mathbf{X}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix}. \quad (22)$$

This first-order form is directly suitable for numerical integration using MATLAB's ordinary differential equation solvers.

7 Simulation Parameters

The baseline parameters used for the numerical simulation are listed in Table 1.

Table 1: Baseline parameters of the quarter-car model.

Parameter	Symbol	Value
Sprung mass	m_s	300 kg
Unsprung mass	m_u	40 kg
Suspension stiffness	k_s	15 000 N/m
Suspension damping	c_s	1500 N s/m
Tyre stiffness	k_t	150 000 N/m
Road amplitude	A	0.05 m
Road wavelength	λ	5 m
Vehicle velocity	v	5 m/s
Excitation frequency	f	1 Hz
Angular frequency	ω	6.283 rad/s
Simulation time	t_{sim}	1 s

8 Characteristic Suspension Frequency

A useful preliminary estimate of the suspension natural frequency can be obtained by considering the sprung mass and suspension stiffness:

$$f_n \approx \frac{1}{2\pi} \sqrt{\frac{k_s}{m_s}}. \quad (23)$$

For the baseline parameters,

$$f_n \approx \frac{1}{2\pi} \sqrt{\frac{15000}{300}} \quad (24)$$

which gives

$$f_n \approx 1.13 \text{ Hz}. \quad (25)$$

This is only an approximate sprung-mass natural frequency. The complete quarter-car model has two coupled dynamic modes because both the sprung and unsprung masses participate in the motion.

9 MATLAB Implementation

The mathematical model was implemented in MATLAB using numerical integration. The ordinary differential equation solver `ode45` was used to obtain the time response of the four state variables.

The main simulation procedure consists of the following steps:

1. Define the physical and road parameters.

2. Calculate the road excitation frequency from vehicle velocity and wavelength.
3. Define the initial conditions.
4. Integrate the governing equations using `ode45`.
5. Extract sprung and unsprung displacement and velocity.
6. Calculate sprung and unsprung acceleration.
7. Calculate suspension and tyre deflections.
8. Determine the maximum response values.
9. Plot the resulting time-domain responses.

9.1 MATLAB Parameter Definition

The baseline parameters were defined using the following MATLAB code:

```
ms = 300;
mu = 40;

ks = 15000;
cs = 1500;
kt = 150000;

A = 0.05;
lambda = 5;

v = 5;

f = v/lambda;
omega = 2*pi*f;

t_sim = 1;
```

9.2 Numerical Solution

The state vector was initialized as

$$\mathbf{X}(0) = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}^T. \quad (26)$$

The equations were then solved numerically using `ode45`. The numerical solution provides

$$x_s(t), \quad \dot{x}_s(t), \quad x_u(t), \quad \dot{x}_u(t).$$

The accelerations were subsequently calculated from the governing equations.

10 Simulink Implementation

An equivalent block-diagram model was developed in MATLAB/Simulink to verify and visualize the mathematical model.

The two acceleration equations are implemented using the following relationships:

$$\ddot{x}_s = \frac{-k_s(x_s - x_u) - c_s(\dot{x}_s - \dot{x}_u)}{m_s}, \quad (27)$$

and

$$\ddot{x}_u = \frac{k_s(x_s - x_u) + c_s(\dot{x}_s - \dot{x}_u) - k_t(x_u - x_r)}{m_u}. \quad (28)$$

Each acceleration is integrated once to obtain velocity and a second time to obtain displacement.

The Simulink implementation therefore contains:

- a sinusoidal road input,
- subtraction blocks for relative displacement,
- subtraction blocks for relative velocity,
- gain blocks representing k_s , c_s , k_t , $1/m_s$, and $1/m_u$,
- summation blocks for force balance,
- integrators for velocity and displacement,
- output blocks for recording the required responses.

The damping coefficient c_s is used in both the sprung-mass and unsprung-mass equations because the same physical damper acts between the two masses.

The sinusoidal road input is implemented using

$$x_r(t) = A \sin(\omega t), \quad (29)$$

where the frequency input of the Sine Wave block is specified in rad/s. Therefore, the value supplied to the block is ω , rather than f .

For example,

v (m/s)	f (Hz)	ω (rad/s)
5	1	2π
10	2	4π
15	3	6π

This relationship ensures that changing vehicle velocity also changes the frequency of road excitation.

11 Performance Parameters

The following quantities are used to evaluate the dynamic response.

11.1 Maximum Body Displacement

The maximum absolute body displacement is

$$X_{s,\max} = \max |x_s(t)|. \quad (30)$$

This quantity represents the maximum vertical motion of the sprung mass relative to the equilibrium position.

11.2 Maximum Wheel Displacement

The maximum wheel displacement is

$$X_{u,\max} = \max |x_u(t)|. \quad (31)$$

This represents the vertical motion of the unsprung mass.

11.3 Maximum Suspension Deflection

The suspension deflection is

$$x_{\text{susp}} = x_s - x_u. \quad (32)$$

Therefore,

$$X_{\text{susp},\max} = \max |x_s - x_u|. \quad (33)$$

11.4 Maximum Tyre Deflection

The tyre deflection is

$$x_{\text{tyre}} = x_u - x_r. \quad (34)$$

Thus,

$$X_{\text{tyre,max}} = \max |x_u - x_r|. \quad (35)$$

12 Baseline Simulation Results

The baseline simulation was performed using

$$m_s = 300 \text{ kg}, \quad m_u = 40 \text{ kg}, \quad k_s = 15\,000 \text{ N/m},$$

$$c_s = 1500 \text{ N s/m}, \quad k_t = 150\,000 \text{ N/m},$$

with

$$A = 0.05 \text{ m}, \quad \lambda = 5 \text{ m}, \quad v = 5 \text{ m/s}.$$

The corresponding road excitation frequency is

$$f = \frac{v}{\lambda} = 1 \text{ Hz}.$$

The baseline simulation provides the time histories of road displacement, wheel displacement, body displacement, body acceleration, suspension deflection, and tyre deflection.

12.1 Overall Displacement Response

Figure 3 compares the road, wheel, and body displacement responses.

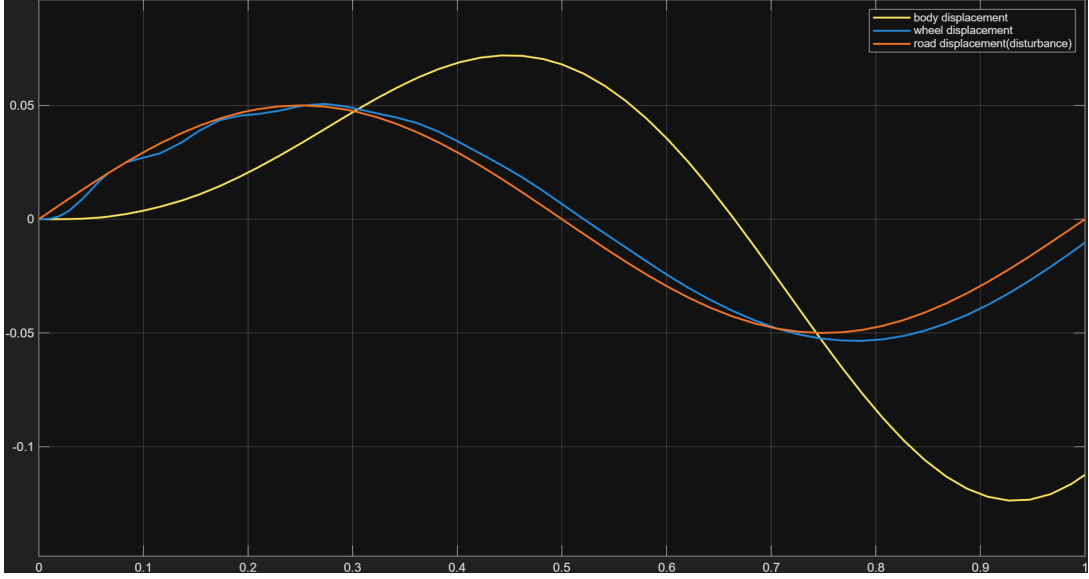


Figure 3: Baseline road, wheel, and body displacement responses.

The road displacement provides the external excitation to the system. The unsprung mass follows the road input with a modified amplitude and phase, while the suspension isolates part of the road motion from the sprung mass. Consequently, the body displacement is different from the road displacement and is governed by the coupled suspension dynamics.

12.2 Body Displacement

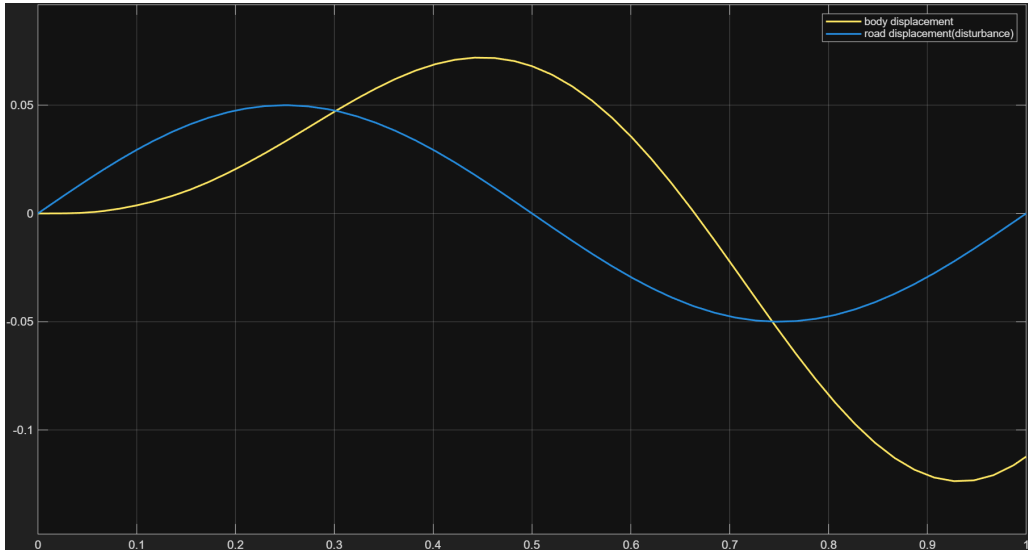


Figure 4: Baseline vehicle body displacement response.

The body displacement represents the vertical movement of the sprung mass. The response depends on the relationship between the road excitation frequency and the dynamic characteristics of the suspension system.

12.3 Suspension Deflection

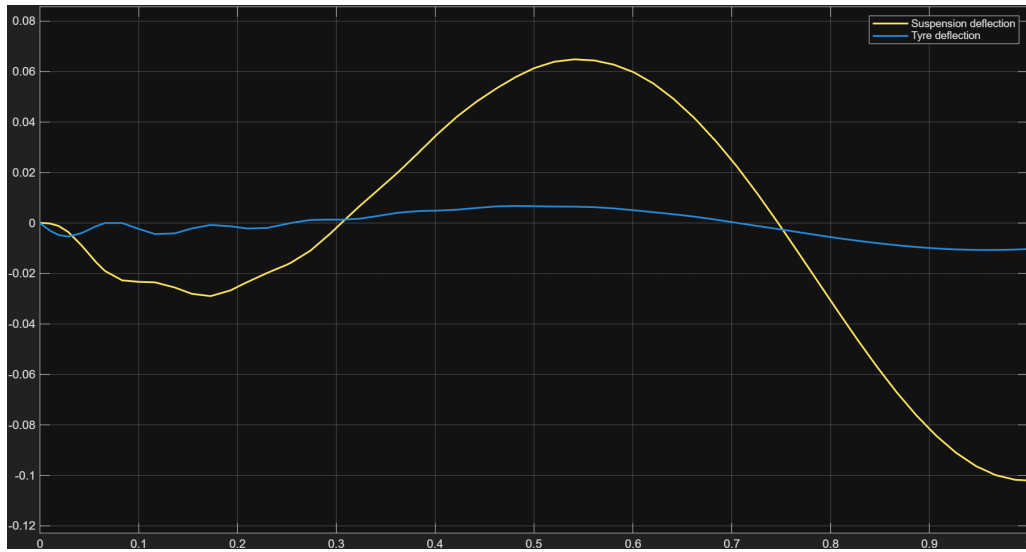


Figure 5: Baseline suspension deflection.

The suspension deflection is the relative displacement between the sprung and unsprung masses. It indicates the amount of suspension travel required to accommodate the road excitation.

12.4 Body–Wheel Displacement

The vertical displacement response of the sprung mass and unsprung mass is shown in Fig. 6. The response illustrates the dynamic motion of the vehicle body and wheel assembly under the prescribed sinusoidal road excitation.

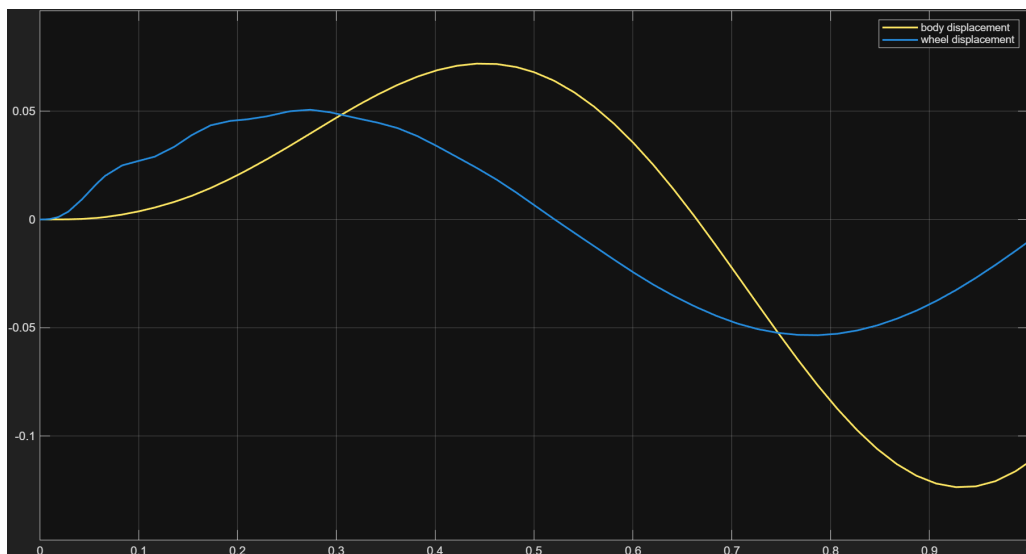


Figure 6: Sprung-mass and unsprung-mass displacement response under baseline road excitation.

12.5 Vehicle Body Displacement and Acceleration Response

The body displacement and acceleration responses under baseline road excitation are shown in Fig. 7.

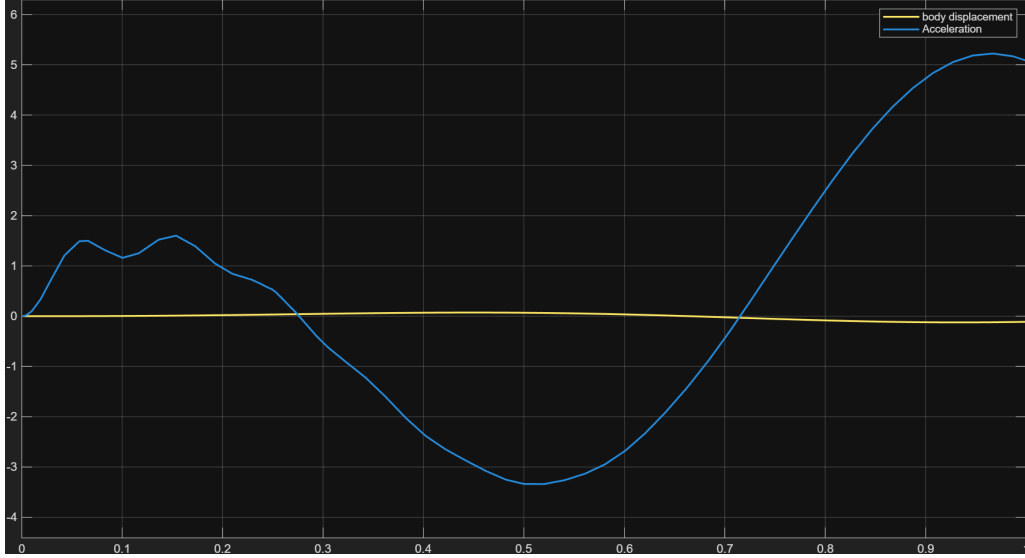


Figure 7: Vehicle body displacement and acceleration response under baseline road excitation.

13 Effect of Suspension Damping

The suspension damping coefficient was varied to investigate its effect on the dynamic response. The vehicle velocity was maintained at

$$v = 5 \text{ m/s},$$

while the damping coefficient was varied as

$$c_s = 500 \text{ N s/m}, \quad 1000 \text{ N s/m}, \quad 1500 \text{ N s/m}.$$

All other model parameters were kept constant.

13.1 Damping Study Results

The maximum response values obtained from the simulations are presented in Table 3.

Table 2: Effect of suspension damping at $v = 5$ m/s.

c_s (N s/m)	Body Displacement (m)	Suspension Deflection (m)	Wheel Displacement (m)	Body Acceleration (m/s ²)
500	0.11230	0.10200	0.01030	5.083
1000	0.08339	0.07593	0.007464	3.686
1500	0.06463	0.05902	0.005608	2.771

Table 3: Effect of suspension damping on the quarter-car response at $v = 5$ m/s.

13.2 Body Displacement with Damping



Figure 8: Body displacement for different suspension damping coefficients.

The maximum body displacement decreases as the damping coefficient is increased for the conditions considered. At 500 N s/m, the maximum body displacement is 0.112 30 m. Increasing the damping coefficient to 1000 N s/m reduces the maximum value to 0.083 39 m, while 1500 N s/m gives a maximum displacement of 0.064 63 m.

The reduction in maximum body displacement between 500 N s/m and 1500 N s/m is

$$\frac{0.11230 - 0.06463}{0.11230} \times 100 \approx 42.45\%. \quad (36)$$

Thus, for the simulated condition, increasing damping substantially reduces the maximum body displacement.

13.3 Suspension Deflection with Damping

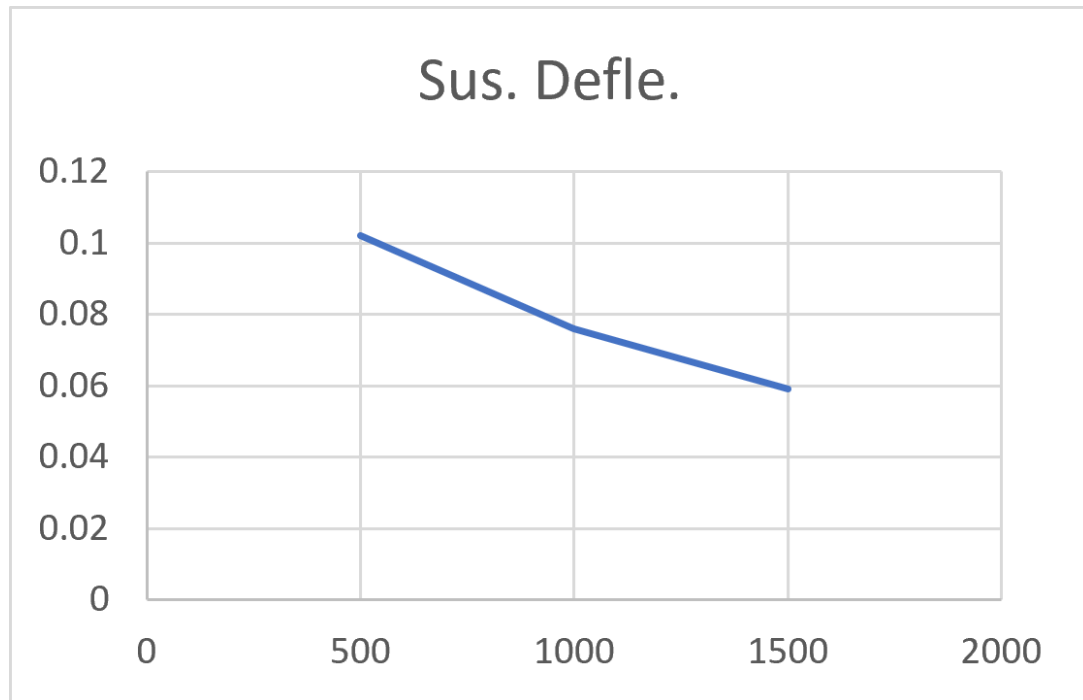


Figure 9: Suspension deflection for different damping coefficients.

The maximum suspension deflection decreases from 0.102 00 m at 500 N s/m to 0.059 02 m at 1500 N s/m. This indicates that the increased damping limits the relative oscillatory motion between the sprung and unsprung masses for the simulated road excitation.

13.4 Wheel Displacement with Damping

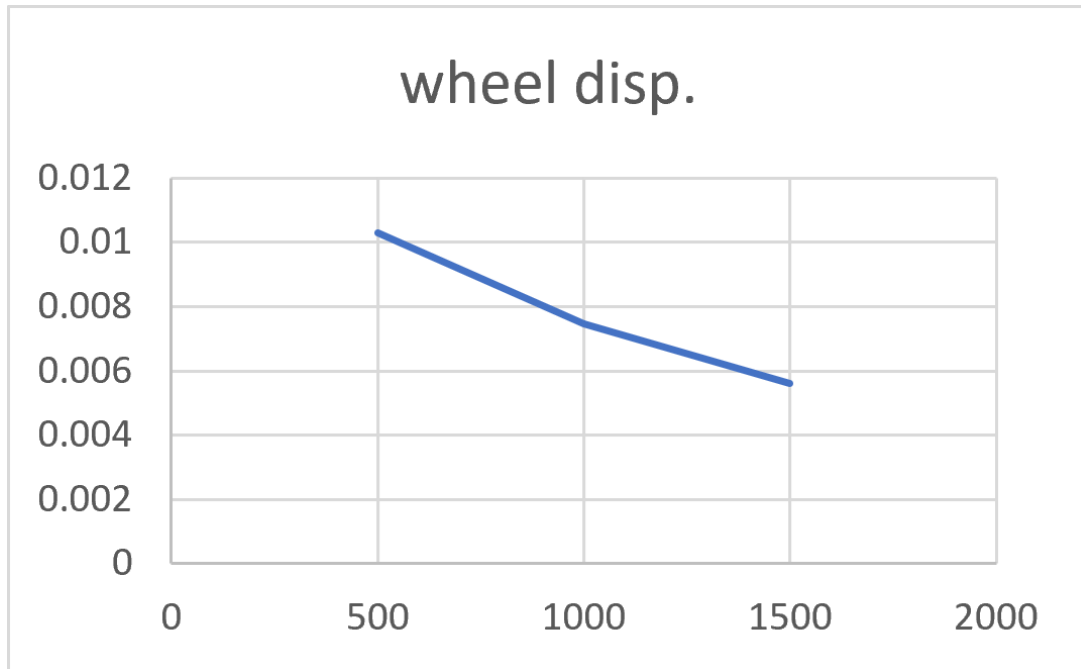


Figure 10: Wheel displacement for different damping coefficients.

The maximum wheel displacement also decreases with increasing damping for the conditions studied. The maximum wheel displacement decreases from 0.010 30 m at 500 N s/m to 0.005 608 m at 1500 N s/m.

13.5 Effect of Damping on Vehicle Body Acceleration

The effect of suspension damping on the vehicle body acceleration response is shown in Fig. 11. The results are compared for damping coefficients of $c_s = 500$, 1000, and 1500 N s/m at a vehicle velocity of 5 m/s.

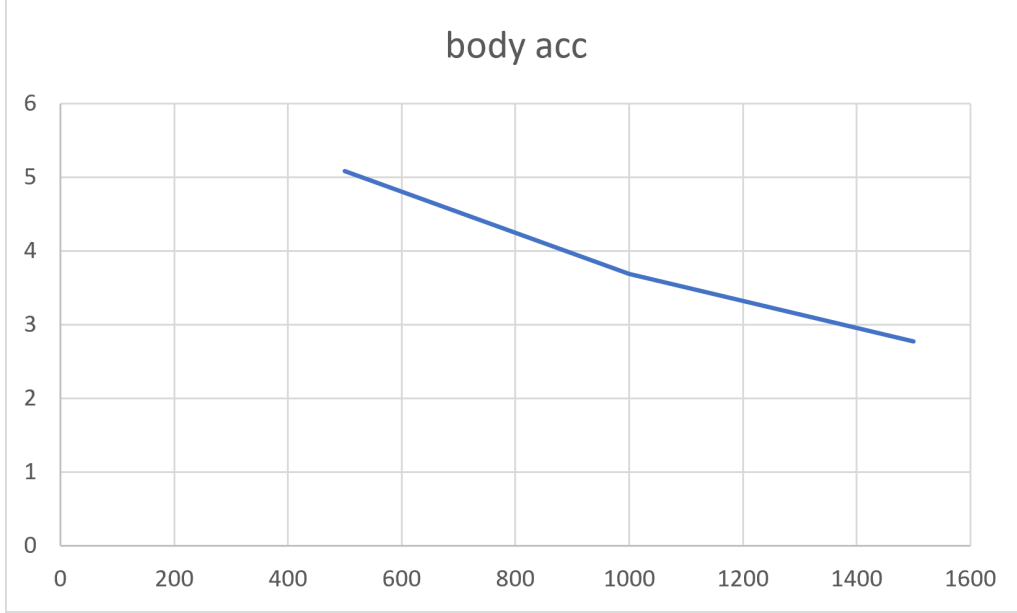


Figure 11: Vehicle body acceleration response for different suspension damping coefficients.

13.6 Interpretation of Damping Results

The damper generates a force proportional to the relative velocity between the sprung and unsprung masses:

$$F_d = c_s(\dot{x}_s - \dot{x}_u).$$

Therefore, increasing c_s increases the dissipative force for a given relative velocity. This reduces the oscillatory response of the suspension system.

For the present simulation conditions, all three measured maximum displacement quantities decrease as the damping coefficient is increased. This trend is specific to the selected model parameters and excitation condition; damping effects in a general suspension system are frequency-dependent and may produce different responses under other operating conditions.

14 Effect of Vehicle Velocity

The effect of vehicle velocity was investigated while maintaining the road amplitude and wavelength constant.

The damping coefficient was fixed at

$$c_s = 1500 \text{ N s/m}.$$

The vehicle velocities considered were

$$v = 5 \text{ m/s}, \quad 10 \text{ m/s}, \quad 15 \text{ m/s}.$$

Since

$$f = \frac{v}{\lambda},$$

and the wavelength was fixed at 5 m, the corresponding excitation frequencies were

v (m/s)	f (Hz)
5	1
10	2
15	3

Thus, increasing vehicle velocity increases the frequency at which the sinusoidal road profile excites the suspension.

14.1 Velocity Study Results

The maximum displacement results are given in Table 5.

Table 4: Effect of vehicle velocity on quarter-car response.

Velocity v (m/s)	Body Displacement (m)	Suspension Deflection (m)	Wheel Displacement (m)	Body Acceleration (m/s ²)
5	0.06463	0.05902	0.005608	2.771
10	0.02589	0.01643	0.009468	4.527
15	0.01332	0.001385	0.01193	5.389

Table 5: Effect of vehicle velocity on the quarter-car suspension response.

14.2 Body Displacement with Velocity

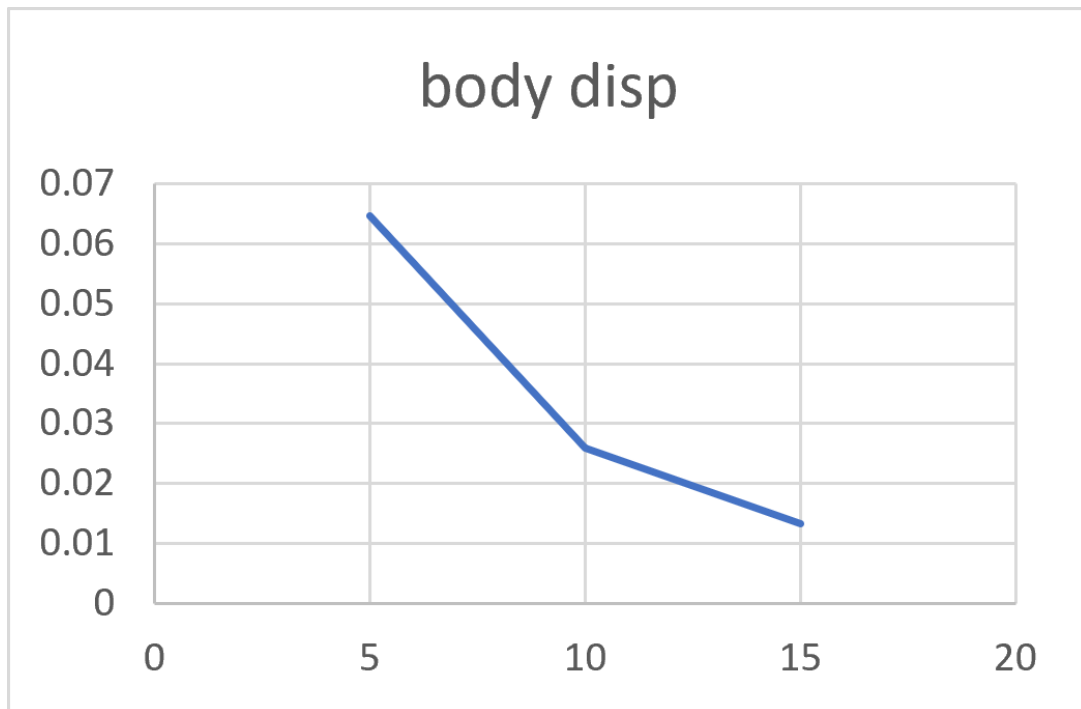


Figure 12: Body displacement for different vehicle velocities.

The maximum body displacement decreases over the velocity range considered. It decreases from 0.064 63 m at 5 m/s to 0.025 89 m at 10 m/s, and further to 0.013 32 m at 15 m/s.

This response is associated with the change in road excitation frequency as the vehicle velocity changes. The result should not be interpreted as a general rule that increasing speed always reduces body displacement. Rather, the response is dependent on the relationship between excitation frequency and the dynamic characteristics of the suspension system.

14.3 Suspension Deflection with Velocity

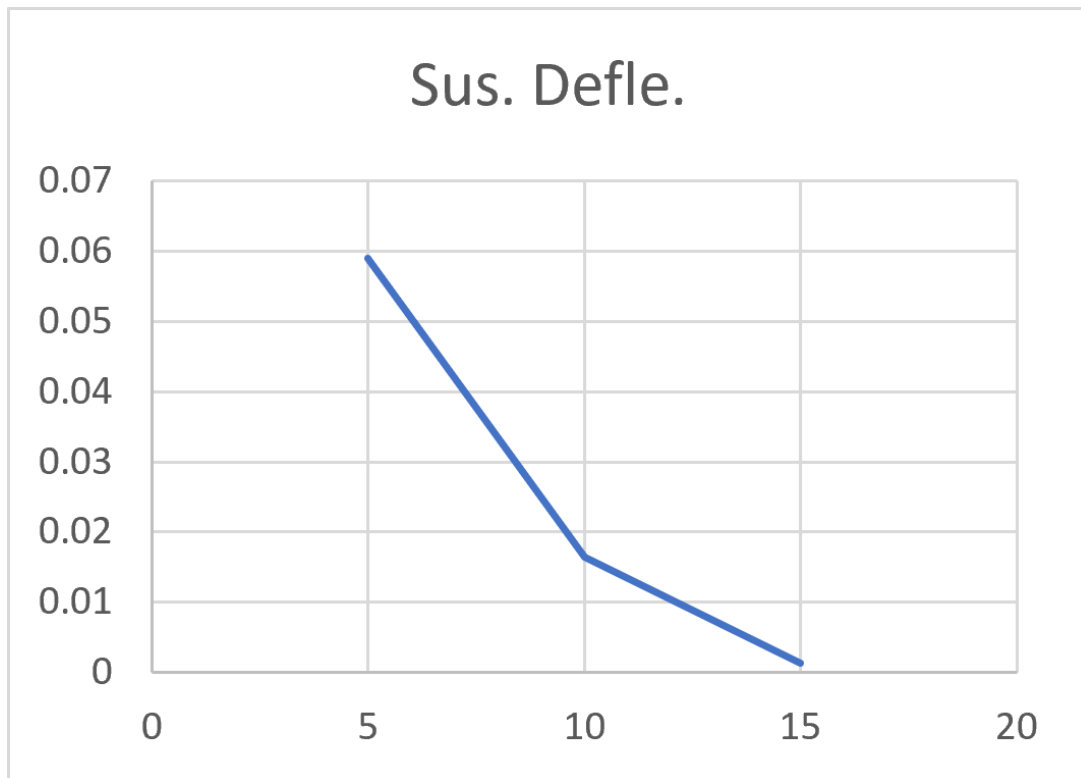


Figure 13: Suspension deflection for different vehicle velocities.

The maximum suspension deflection decreases from 0.059 02 m at 5 m/s to 0.016 43 m at 10 m/s. At 15 m/s, the maximum suspension deflection is 0.001 385 m.

The reduction indicates that, for the selected sinusoidal road profile and model parameters, the relative motion between the sprung and unsprung masses becomes smaller as the excitation frequency is increased over the investigated range.

14.4 Wheel Displacement with Velocity

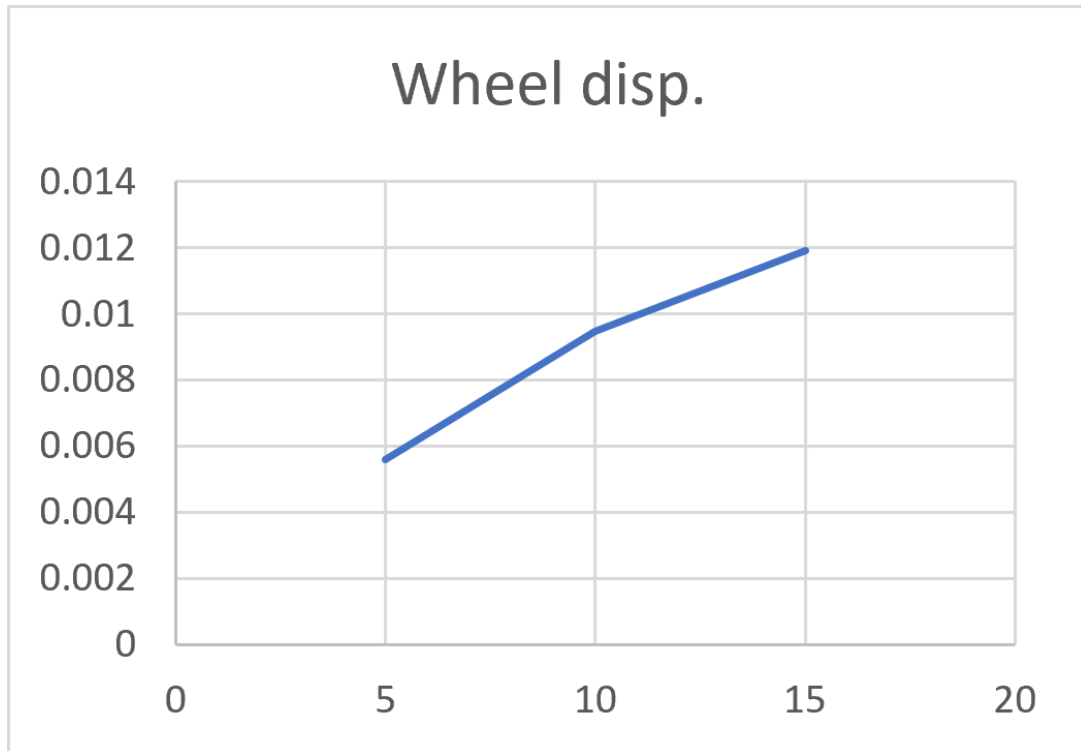


Figure 14: Wheel displacement for different vehicle velocities.

Unlike the body displacement and suspension deflection, the maximum wheel displacement increases with vehicle velocity. It increases from 0.005 608 m at 5 m/s to 0.009 468 m at 10 m/s, and then to 0.011 93 m at 15 m/s.

This difference between body and wheel response demonstrates the coupled nature of the quarter-car system. The sprung and unsprung masses do not necessarily exhibit the same trend when the excitation frequency is changed.

14.5 Effect of Vehicle Velocity on Maximum Body Acceleration

The effect of vehicle velocity on the maximum absolute acceleration of the sprung mass is shown in Fig. 15. The suspension damping coefficient is maintained at $c_s = 1500$ N s/m, while the vehicle velocity is varied from 5 to 15 m/s.

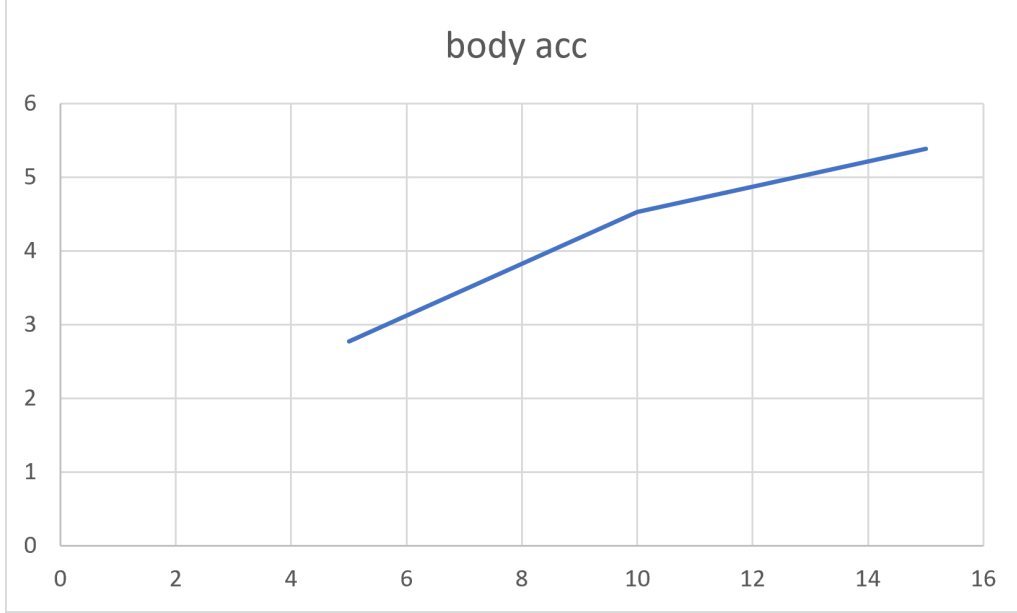


Figure 15: Maximum vehicle body acceleration for different vehicle velocities.

The maximum body acceleration increases with vehicle velocity in the present simulation. The maximum acceleration increases from 2.771 m/s^2 at 5 m/s to 4.527 m/s^2 at 10 m/s and 5.389 m/s^2 at 15 m/s . This occurs because increasing vehicle velocity increases the road excitation frequency according to

$$f = \frac{v}{\lambda}. \quad (37)$$

Thus, the suspension is subjected to a higher-frequency road input as vehicle velocity increases.

15 Comparative Discussion

The simulation results demonstrate that the dynamic response of the quarter-car model depends strongly on both suspension damping and road excitation frequency.

For the damping study, increasing the damping coefficient from 500 N s/m to 1500 N s/m reduces the maximum body displacement from 0.11230 m to 0.06463 m . The maximum suspension deflection decreases from 0.10200 m to 0.05902 m , while the maximum wheel displacement decreases from 0.01030 m to 0.005608 m .

The damping coefficient controls the rate at which mechanical energy is dissipated from the suspension system. Therefore, the larger damping coefficient produces a more strongly dissipative system under the investigated excitation condition.

The velocity study produces a different type of response. Increasing velocity changes the road excitation frequency according to

$$f = \frac{v}{\lambda}.$$

At 5 m/s, the excitation frequency is 1 Hz; at 10 m/s, it is 2 Hz; and at 15 m/s, it is 3 Hz.

The maximum body displacement and suspension deflection decrease over this velocity range, while the maximum wheel displacement increases. This indicates that the sprung and unsprung masses respond differently to changes in excitation frequency.

The results also demonstrate why a quarter-car suspension cannot be characterized using only one output quantity. Body displacement, wheel displacement, suspension travel, tyre deflection, and body acceleration represent different aspects of the dynamic behaviour and should be considered together when evaluating suspension performance.

16 Model Limitations

Although the quarter-car model is useful for studying fundamental suspension behaviour, several limitations should be recognized.

First, the model considers only vertical motion and therefore does not capture vehicle pitch, roll, yaw, or lateral dynamics. Second, the suspension components are represented by linear spring and damper elements, whereas real suspension systems can exhibit non-linear characteristics.

The tyre is represented only by a linear vertical stiffness and tyre damping is neglected. Real tyres exhibit nonlinear stiffness, damping, contact-patch effects, and potentially complex transient behaviour.

The road profile is also idealized as a sinusoidal input. Actual roads contain random, transient, and spatially varying irregularities. Therefore, the response obtained in this study represents the behaviour of the model under a controlled periodic excitation rather than a complete representation of real-road operation.

Finally, the quarter-car model treats one wheel independently and does not account for coupling between the four wheels through the vehicle body.

17 Conclusion

A two-degree-of-freedom quarter-car suspension model was developed to investigate the vertical dynamic response of a vehicle subjected to sinusoidal road excitation. The model consists of a sprung mass, unsprung mass, suspension spring, suspension damper, and tyre stiffness.

The governing equations were derived using Newton's second law and implemented in MATLAB using numerical integration. An equivalent Simulink block diagram was also

developed to represent the same dynamic system.

The damping study showed that increasing the suspension damping coefficient from 500 Ns/m to 1500 Ns/m reduced the maximum body displacement from 0.112 30 m to 0.064 63 m, corresponding to an approximately 42.45% reduction under the specified simulation condition. Maximum suspension deflection and wheel displacement also decreased over the same damping range.

The velocity study demonstrated that vehicle velocity changes the frequency of road excitation according to

$$f = \frac{v}{\lambda}.$$

For the investigated velocities of 5 m/s, 10 m/s, and 15 m/s, the maximum body displacement decreased from 0.064 63 m to 0.013 32 m, while the maximum wheel displacement increased from 0.005 608 m to 0.011 93 m. The maximum suspension deflection also decreased over the investigated range.

Overall, the results demonstrate that suspension damping and excitation frequency significantly influence the dynamic behaviour of the sprung and unsprung masses. The quarter-car model provides a useful foundation for understanding suspension dynamics and for conducting preliminary parameter studies before progressing to more detailed vehicle models.

18 Future Scope

The present model can be extended in several directions to obtain a more realistic representation of vehicle suspension behaviour.

1. A nonlinear suspension spring can be introduced to represent progressive or nonlinear stiffness characteristics.
2. A nonlinear damper model can be implemented to account for different compression and rebound damping characteristics.
3. Tyre damping can be incorporated into the model.
4. A half-car model can be developed to include vehicle pitch dynamics.
5. A full-car model can be developed to include pitch, roll, and four-wheel interactions.
6. Real measured road profiles can replace the sinusoidal road excitation.
7. Random road excitation based on road roughness classifications can be investigated.
8. Suspension parameters can be optimized using numerical optimization techniques.

9. Body acceleration can be evaluated using ride-comfort criteria.
10. Suspension travel and tyre load variation can be investigated as additional performance measures.
11. The model can be extended to study active and semi-active suspension systems.
12. Experimental validation can be performed using a physical quarter-car test rig or vehicle measurement data.

19 References

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